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(54) **INSERTION DEVICE POSITIONING
GUIDANCE SYSTEM AND METHOD**

(71) Applicant: **ENVUE MEDICAL HOLDINGS,
CORP.**, Arlington Heights, IL (US)

(72) Inventors: **Ran Cohen**, Petah-Tikva (IL); **Amit
Cohen**, Petah-Tikva (IL); **Doron
Besser**, Tel Aviv (IL)

(73) Assignee: **ENVUE MEDICAL HOLDINGS,
CORP.**, Arlington Heights, IL (US)

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(2016.02); **A61J 15/0084** (2015.05); **A61B**
2034/2051 (2016.02); **A61M 2202/0482**
(2013.01)

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CPC A61J 15/0015; A61J 15/0084; A61J
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(56) **References Cited**

U.S. PATENT DOCUMENTS

5,558,091 A 9/1996 Acker et al.
6,373,240 B1 4/2002 Govari
(Continued)

FOREIGN PATENT DOCUMENTS

CN 102316799 1/2012
CN 102999902 3/2013
(Continued)

Primary Examiner — Joel Lamprecht

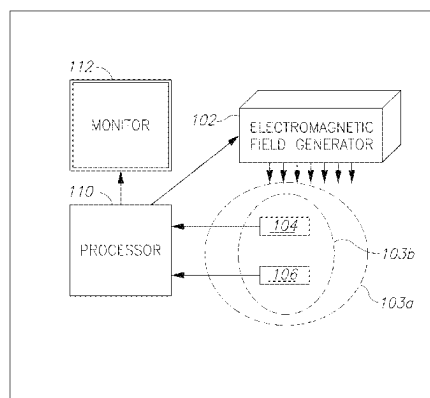
(74) *Attorney, Agent, or Firm* — The Roy Gross Law
Firm, LLC; Roy D. Gross

(57) **ABSTRACT**

There is provided herein a system and a method for guiding
insertion of a gastrointestinal tube including: an electromag-
netic field generator configured to generate an electromag-
netic field covering a treatment area; wherein said electro-
magnetic field generator is external to the patient; a regis-
tration sensor configured to mark anatomic locations on
the patient's torso; a gastrointestinal tube comprising a tip
sensor configured to sense its position and/or orientation
relative to the electromagnetic field generator; and a pro-
cessing circuitry configured to: calculate an orientation of
the subject relative to the field generator based on the
anatomic locations marked by the registration sensor, load a
predefined anatomic map representing a torso; aligning the
map based on the anatomic locations marked by the regis-
tration sensor, and showing on the map a path of the
gastrointestinal tube insertion; wherein the path is generated
according to changes in the strength of the electromagnetic
field sensed by the tip sensor's during the insertion of the
gastrointestinal tube, independent of the subject's movement
and independent of deviations in the position and/or orien-
tation of said field generator.

7 Claims, 10 Drawing Sheets

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continuation of application No. 16/386,493, filed on Apr. 17, 2019, now Pat. No. 10,548,815.

- (60) Provisional application No. 62/664,447, filed on Apr. 30, 2018.

(58) **Field of Classification Search**

CPC A61B 2034/2051; A61B 5/687; A61B 5/6871; A61B 34/25; A61B 5/065; A61B 2034/2068; A61B 2034/2072; A61B 5/062; A61M 2202/0482

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,650,927	B1	11/2003	Keidar
6,690,963	B2	2/2004	Ben-Haim et al.
7,397,364	B2	7/2008	Govari
8,046,050	B2	10/2011	Govari et al.
8,934,960	B2	1/2015	Besz et al.
10,010,374	B2	7/2018	Besser et al.
10,548,815	B2 *	2/2020	Cohen A61B 5/062
11,364,179	B2 *	6/2022	Cohen A61B 34/20
2005/0004456	A1	1/2005	Thomas et al.
2005/0281385	A1	12/2005	Johnson et al.
2006/0030771	A1	2/2006	Levine et al.
2006/0055712	A1	3/2006	Anderson
2008/0228066	A1	9/2008	Waitzman et al.
2009/0234224	A1	9/2009	Justin et al.
2010/0097373	A1	4/2010	Besz et al.

2011/0160569	A1	6/2011	Cohen et al.
2011/0237933	A1	9/2011	Cohen
2013/0046172	A1	2/2013	Waitzman et al.
2014/0243614	A1	8/2014	Rothberg et al.
2014/0354300	A1	12/2014	Ramachandran et al.
2015/0272694	A1	10/2015	Charles
2016/0258782	A1	9/2016	Sadjadi et al.
2016/0354012	A1	12/2016	Zeng et al.
2017/0100055	A1	4/2017	Cronin et al.
2017/0128141	A1	5/2017	Schneider et al.
2018/0049810	A1	2/2018	Besser et al.
2018/0280093	A1	10/2018	Besser et al.
2018/0368716	A1	12/2018	Govari et al.
2019/0328620	A1	10/2019	Cohen et al.
2020/0319267	A1	10/2020	Ashe
2021/0137608	A1	5/2021	Besser et al.
2021/0282717	A1	9/2021	Loring et al.

FOREIGN PATENT DOCUMENTS

CN	107754069	3/2018
CN	107970074	5/2018
EP	1374791	1/2004
EP	3563762	11/2019
JP	2001-524012	11/2001
JP	2014-515628	7/2014
JP	2015-502790	1/2015
JP	2018-27308	2/2018
WO	9849928	11/1998
WO	2007025081	3/2007
WO	2012122002	9/2012
WO	2013078348	5/2013

* cited by examiner

100

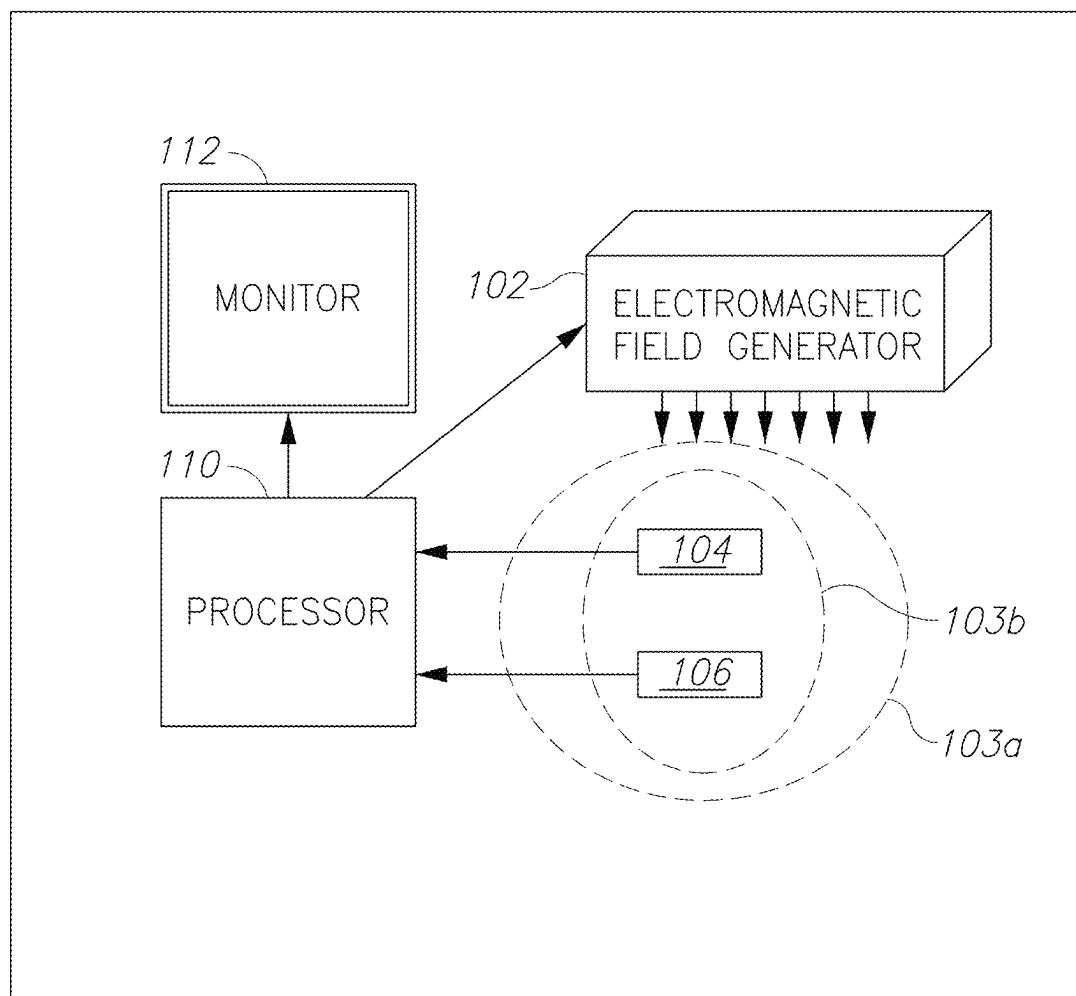


FIG.1

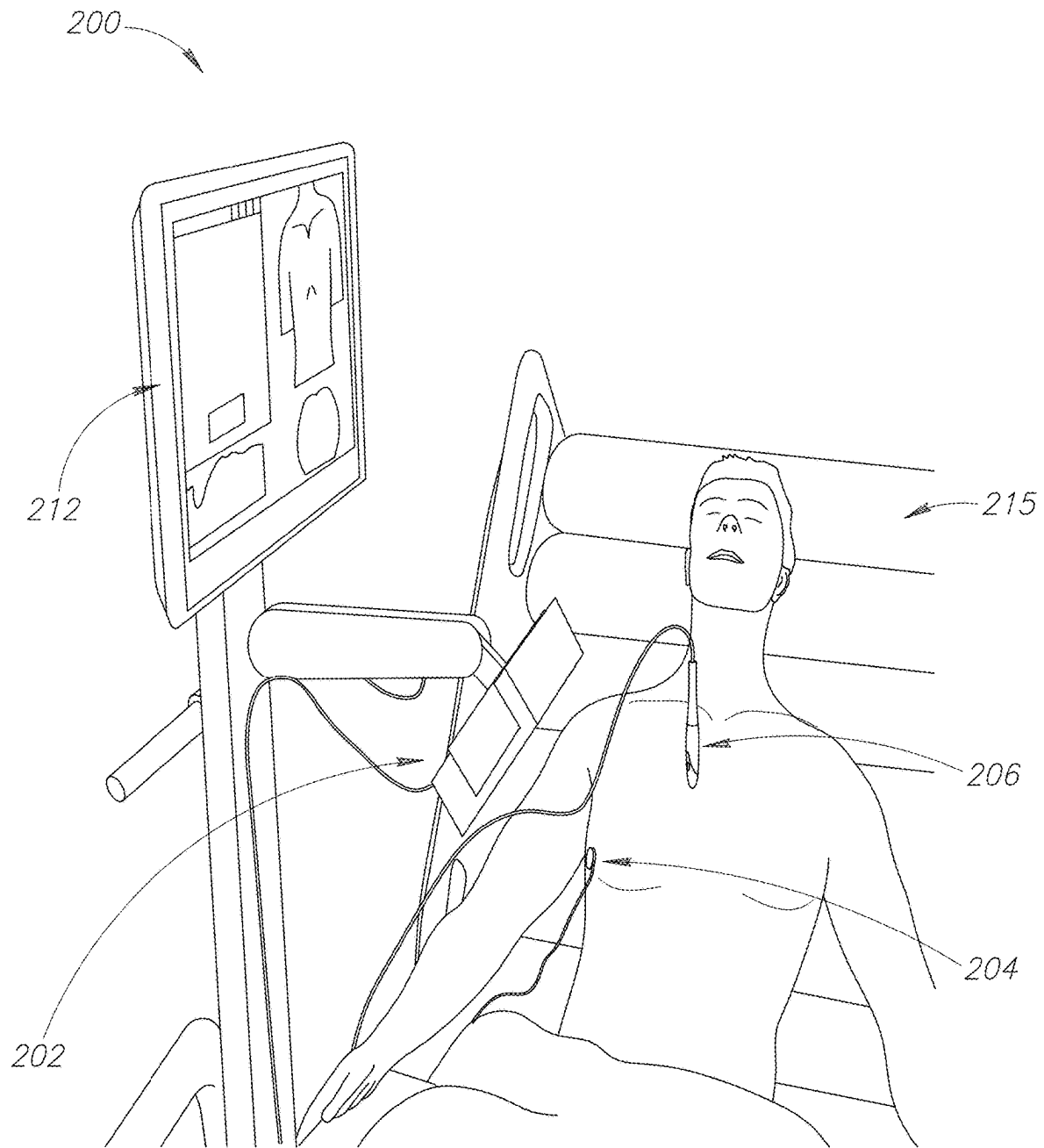


FIG. 2A

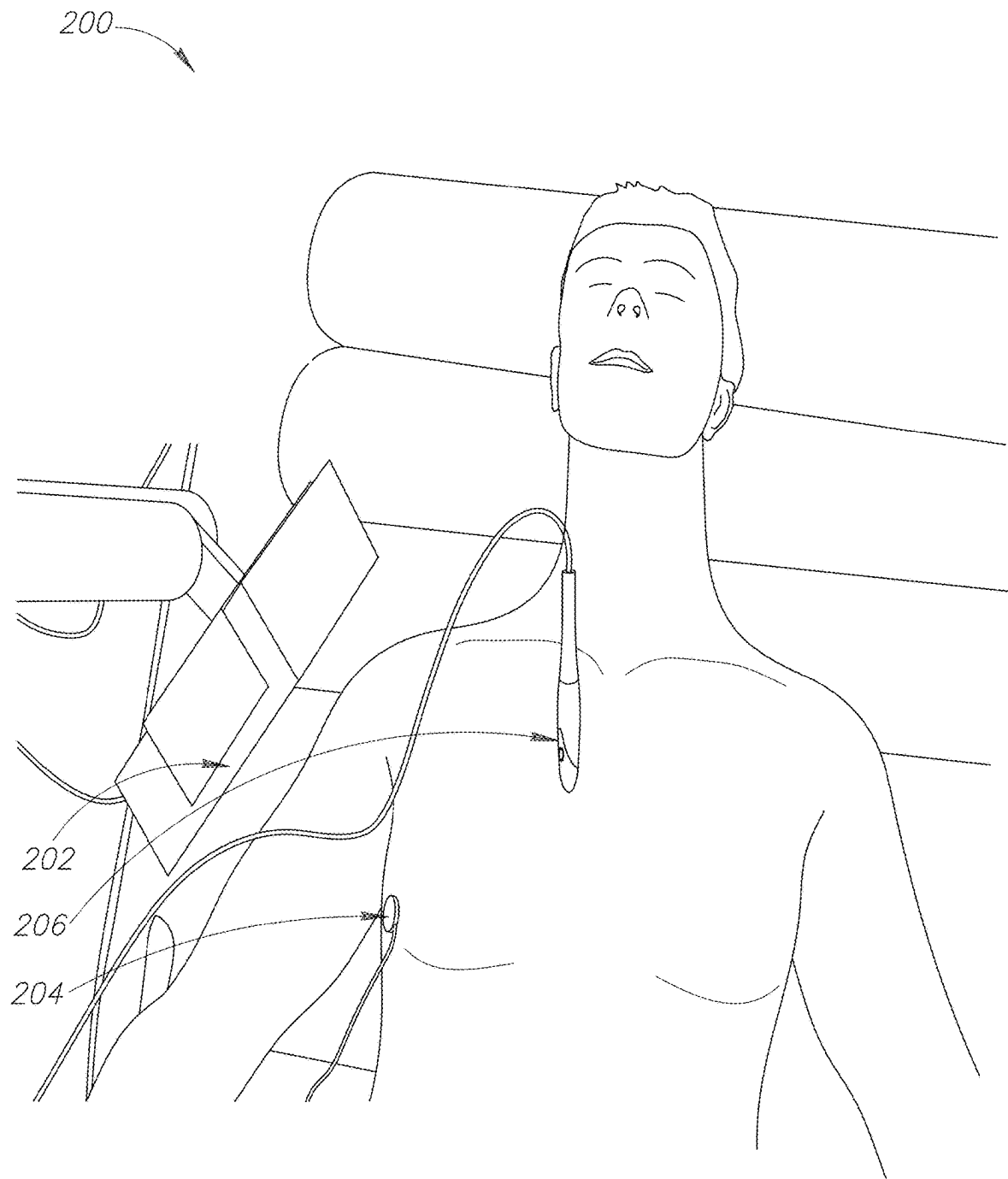


FIG. 2B

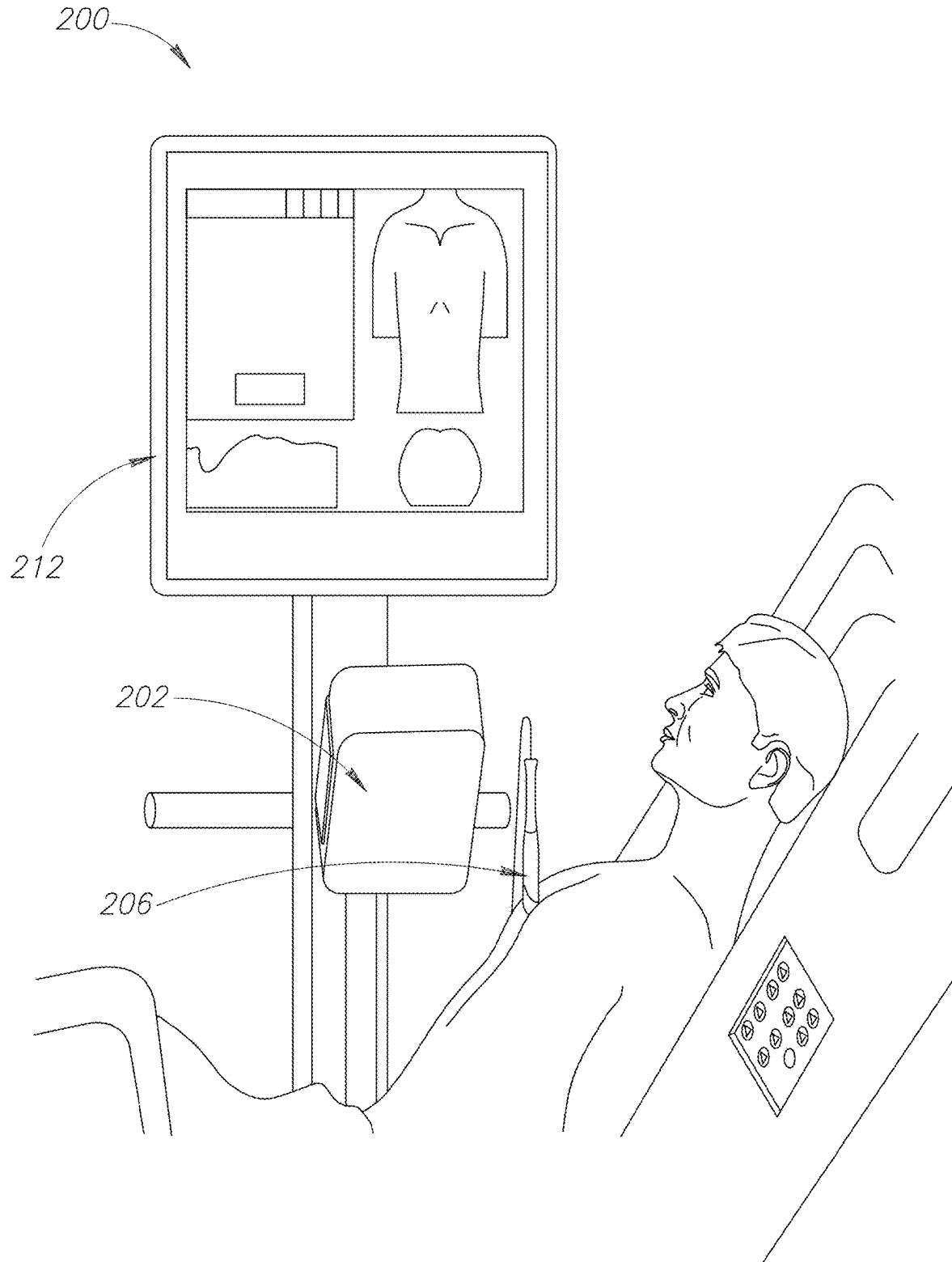


FIG. 2C

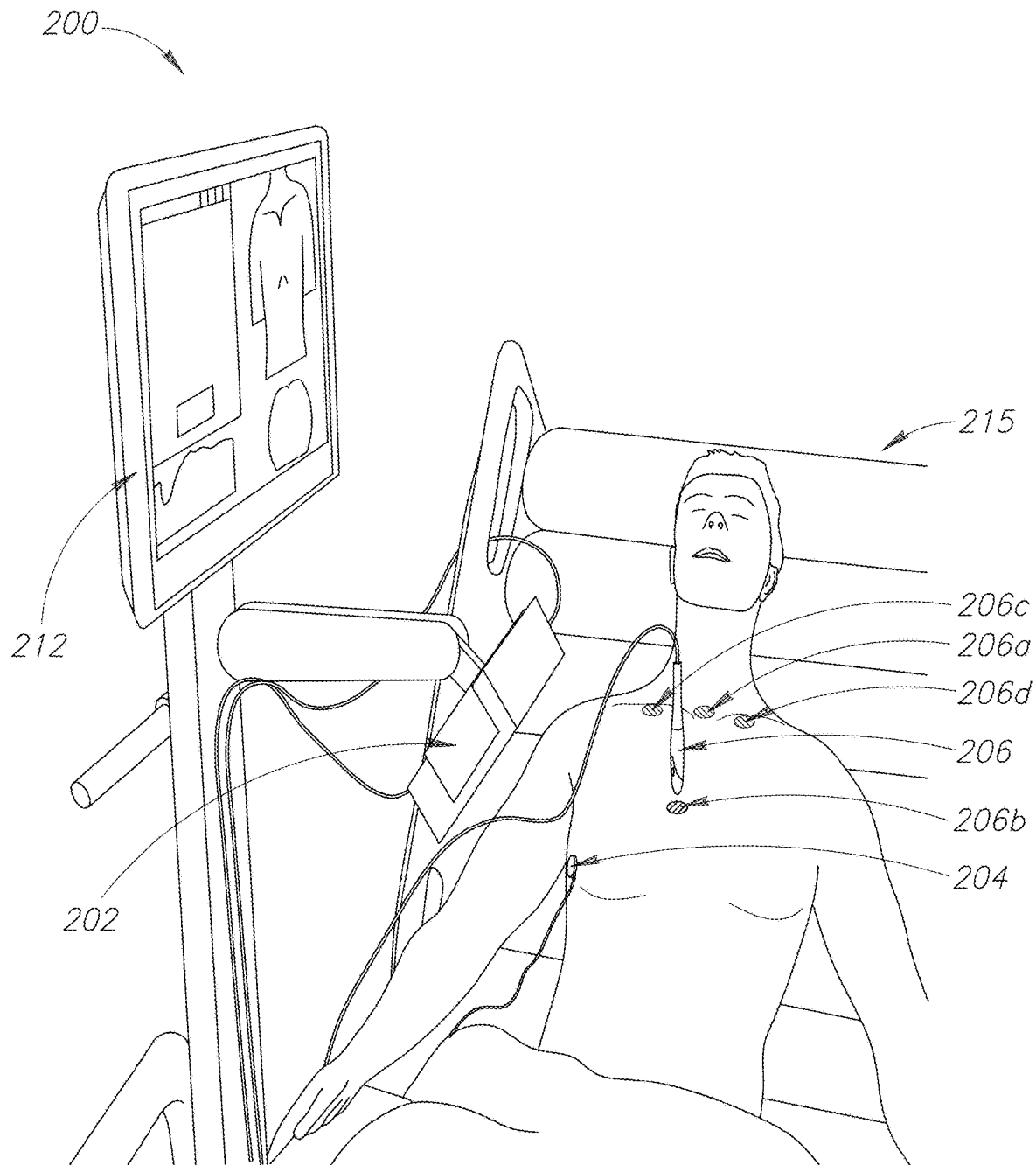


FIG. 2D

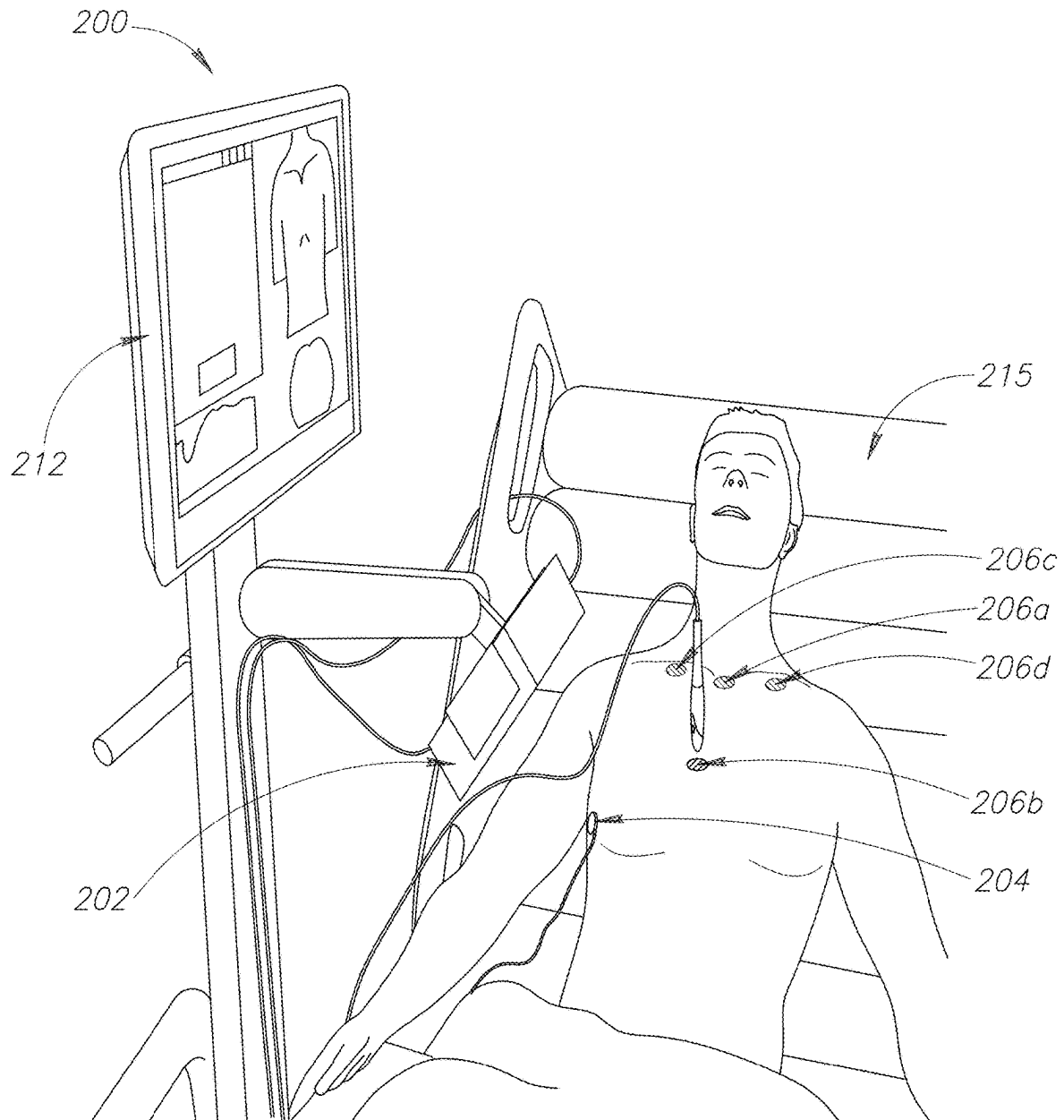


FIG. 2E

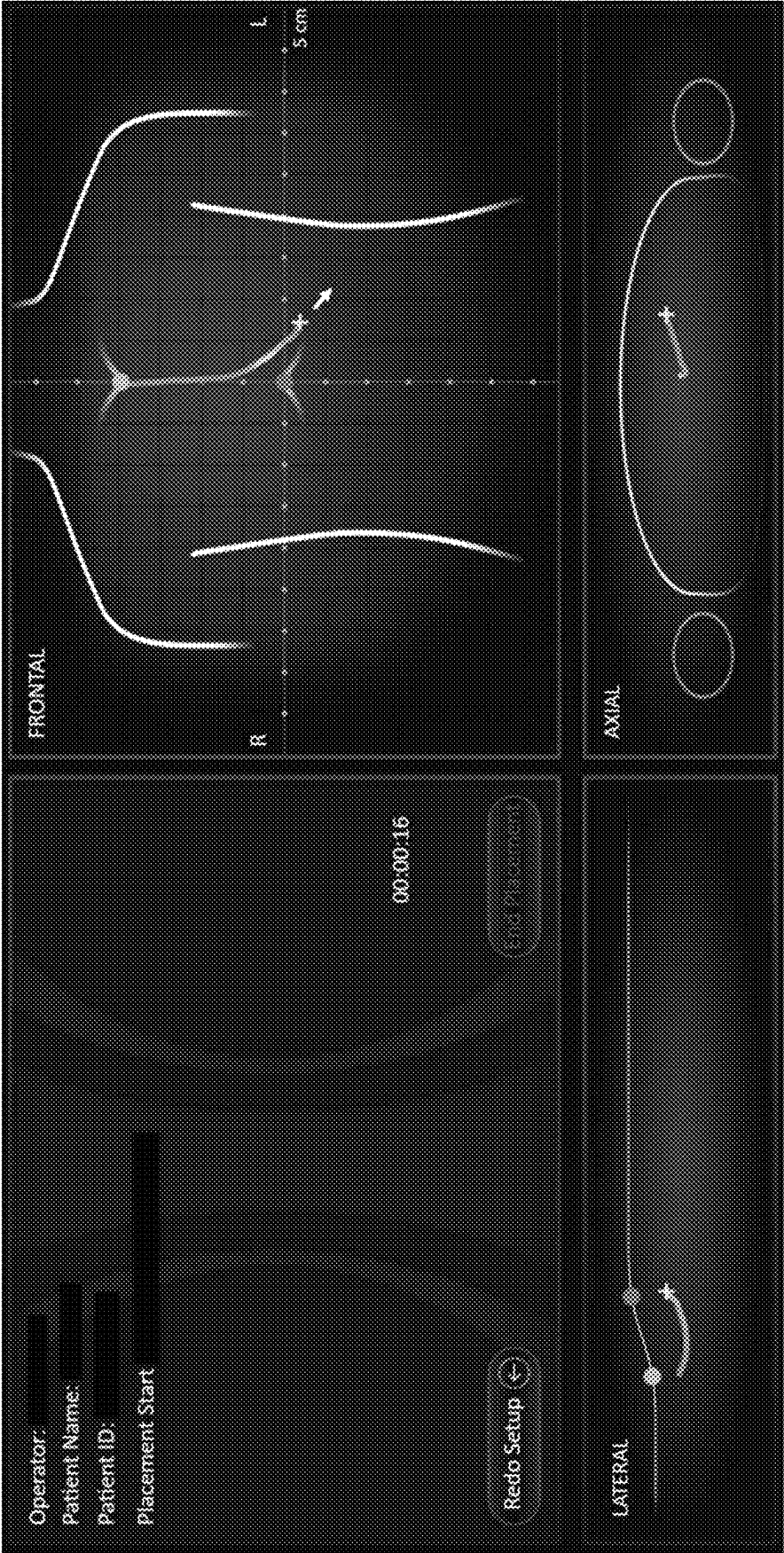


FIG.3A

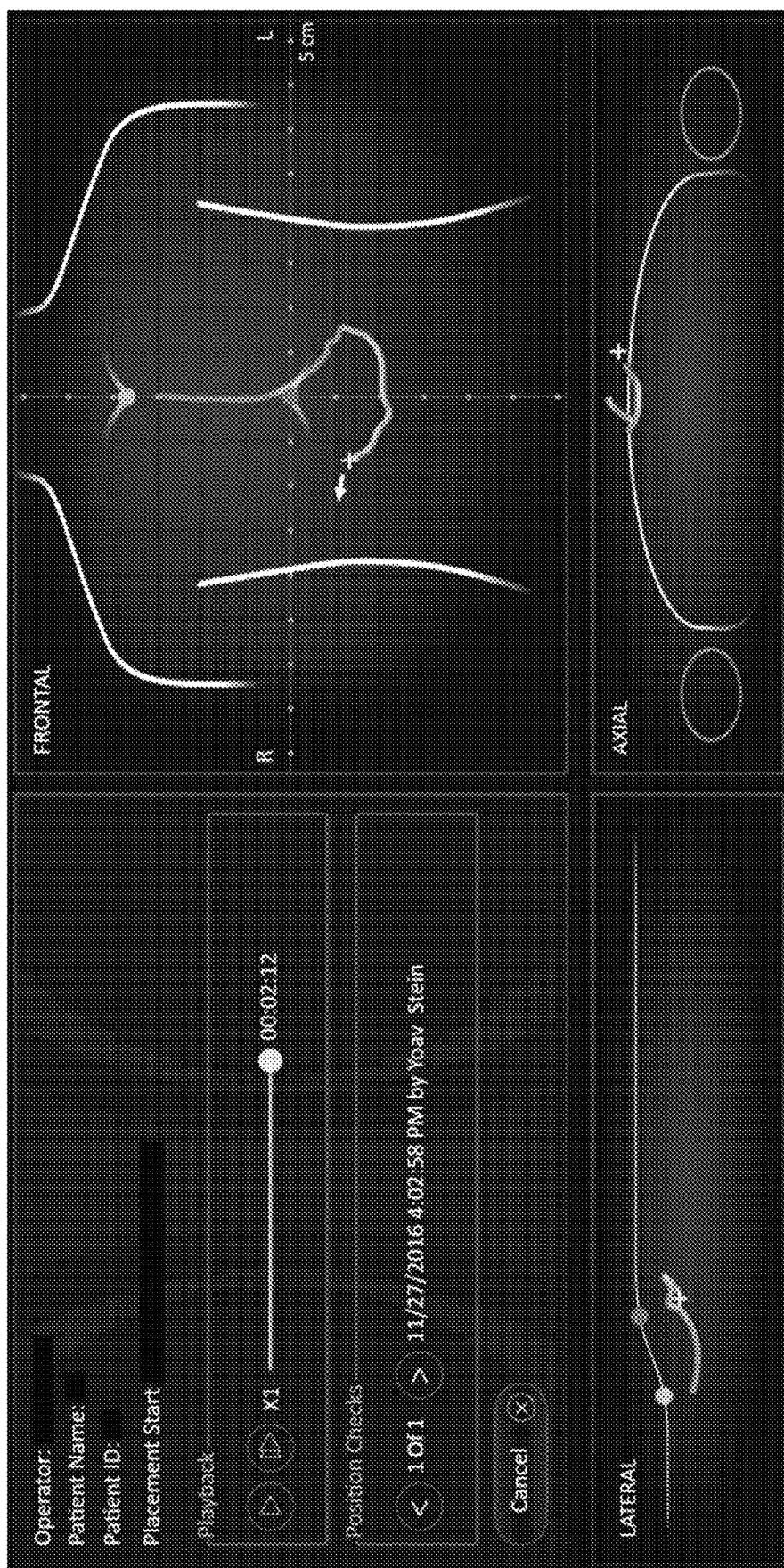


FIG. 3B

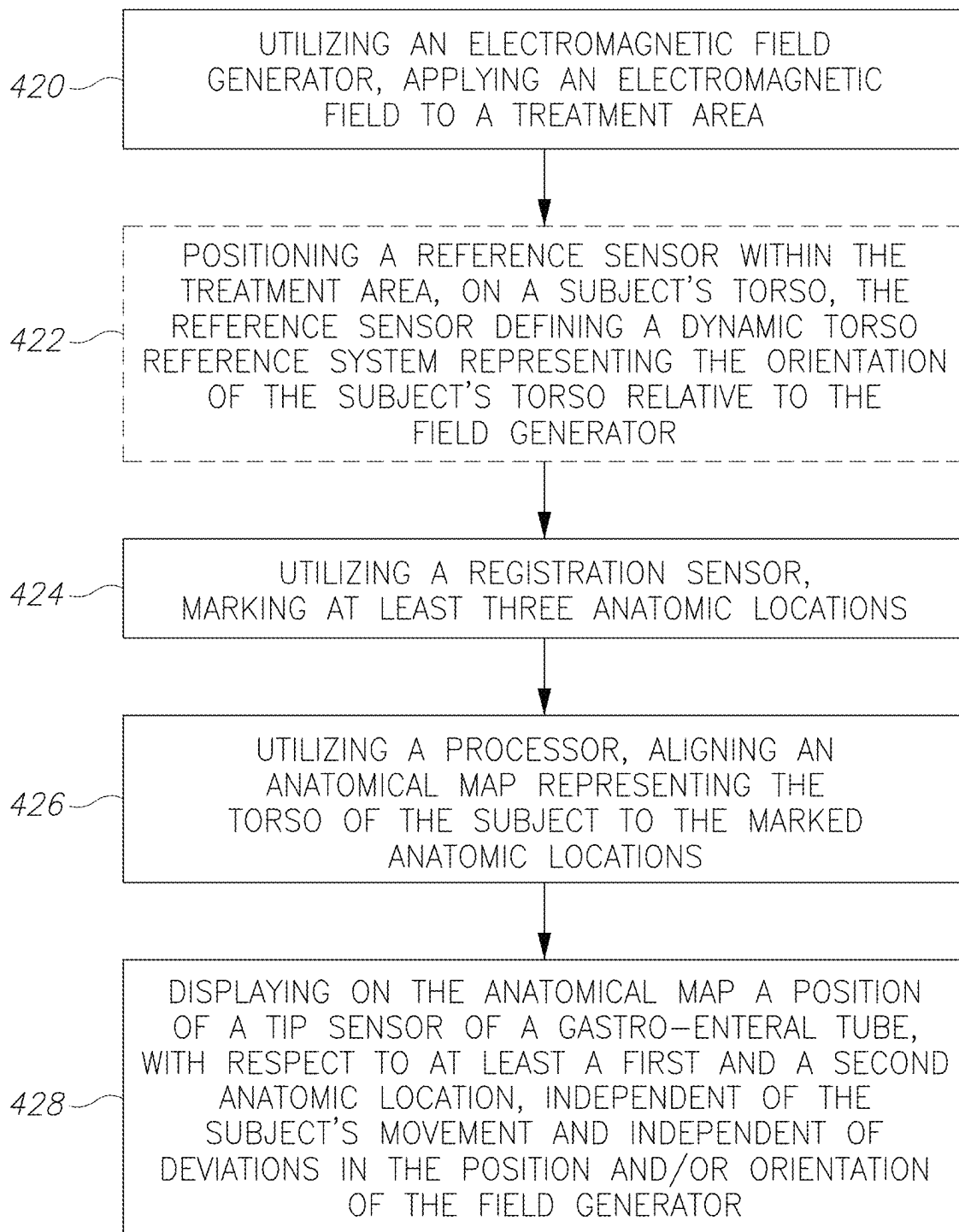


FIG.4

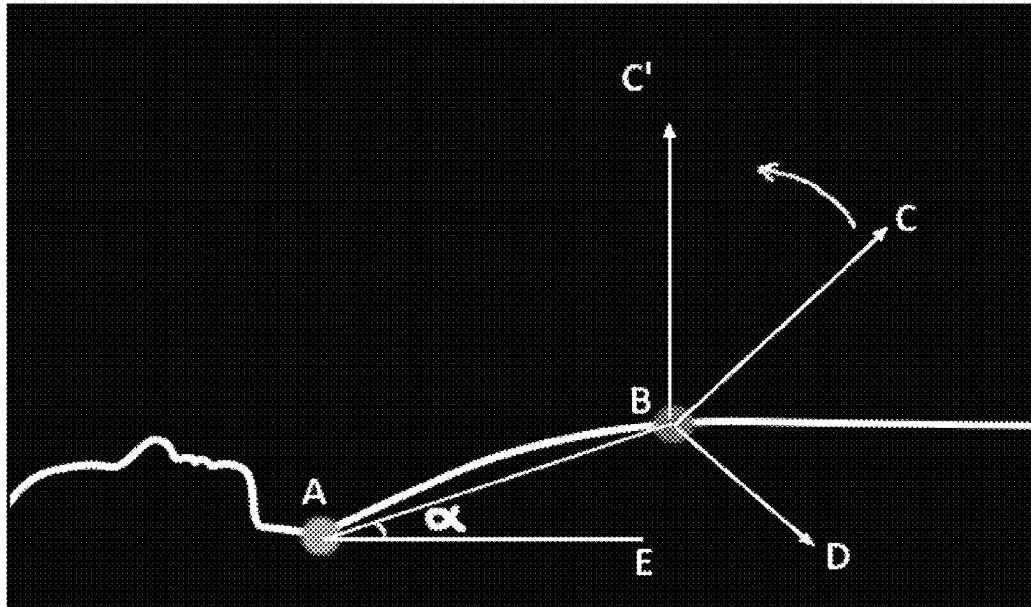


FIG. 5

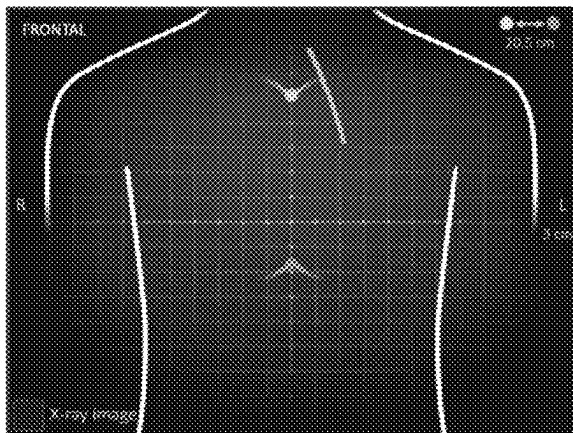


FIG. 6A

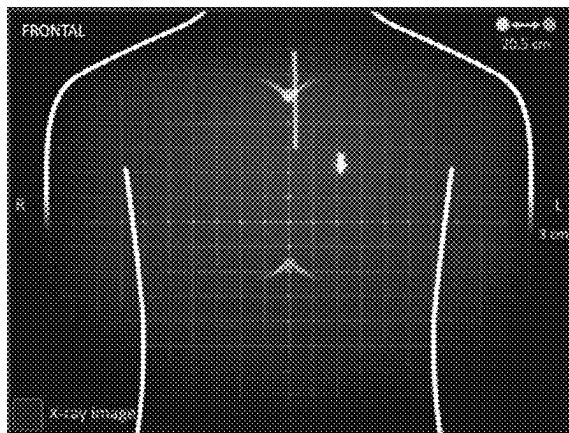


FIG. 6B

1

INSERTION DEVICE POSITIONING GUIDANCE SYSTEM AND METHOD

CROSS REFERENCE TO RELATED APPLICATIONS

This application is continuation of U.S. application Ser. No. 16/724,323 filed on Dec. 22, 2019 (issued as U.S. Pat. No. 11,364,179), which is a continuation of U.S. application Ser. No. 16/386,493 filed on Apr. 17, 2019, now U.S. Pat. No. 10,548,815, issued on Feb. 4, 2020 that claims the benefit of priority of U.S. Provisional Application No. 62/664,447 filed on Apr. 30, 2018. The contents of the above application are incorporated by reference as if fully set forth herein in its entirety.

FIELD OF INVENTION

Embodiments of the disclosure relate to insertion device positioning guidance systems and methods.

BACKGROUND OF THE INVENTION

Enteral feeding is often used as nutritional support in patients unable to be fed otherwise. Although many benefits are associated with early initiation of enteral feeding, misplacement of feeding tubes is relatively common and can result in patient discomfort and complications. Confirming the position of the tube only after it is already inserted delays the feeding and the initiating of hydration or medication. Bedside electromagnetic (EM) systems for guided placement of nasogastric feeding tubes are available and are utilized by medical staff during the procedure to avoid misplacement of feeding tubes. There is still a need, however, for reliable real-time tracking systems that provide enhanced accuracy for critical tool positioning during medical procedures.

The foregoing examples of the related art and limitations related therewith are intended to be illustrative and not exclusive. Other limitations of the related art will become apparent to those of skill in the art upon a reading of the specification and a study of the figures.

SUMMARY OF THE INVENTION

The following embodiments and aspects thereof are described and illustrated in conjunction with systems, tools and methods, which are meant to be exemplary and illustrative, not limiting in scope.

One of the problems often associated with an insertion of a feeding tube using an electromagnetic positioning guidance system, is that reliability is difficult to obtain in the patient environment, which is typically dynamic. For example, the patient often moves, and the bed is moved from one place to another. There is thus provided herein an electromagnetic positioning guidance system reliably operable regardless of the patient's movement or position and which requires no calibration.

According to some embodiments, there is provided an insertion device positioning guidance system comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; wherein said electromagnetic field generator is external to the patient; a registration sensor configured to mark at least three anatomic locations on the patient's torso; a gastroenteral tube comprising a tip sensor configured to sense its location relative to the electromagnetic field generator; and

2

a processing circuitry configured to: calculate an orientation of the subject relative to the field generator based on the three anatomic locations marked by the registration sensor, load a predefined anatomical map representing a torso; aligning the map based on positions corresponding to or adjacent to the suprasternal notch and an atomic location below the suprasternal notch, such as, but not limited to, the xiphoid process, based on the at least three anatomic locations marked by the registration sensor, and showing on the map a path of the gastroenteral tube insertion; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the gastroenteral tube, while minimizing sensitivity to the subject's movement and/or deviations in the position and/or orientation of the field generator.

According to some embodiments, the system further comprises a reference sensor configured to be positioned, within the treatment area, on the subject's torso, the reference sensor configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator;

In some embodiments, the reference sensor is configured to be positioned on a side of the patient's torso. In some embodiments, the anatomical map shows a frontal upper view of the subject essentially parallel to the patient's posture. In some embodiments, the anatomical map shows a side view of the subject. In some embodiments, the anatomical map shows an axial view of the subject.

In some embodiments, the system further comprises a monitor configured to display the anatomical map.

According to some embodiments, the term "anatomical map" as disclosed herein, may refer to one or more schematic maps, one or more 2D anatomical maps, one or more 3D anatomical maps, or any combination thereof. According to some embodiments, the term "anatomical map" as disclosed herein, may refer to a group of maps (e.g., 2, 3, 4 or more), each representing a different view (for example, frontal view, frontal upper view, side view, axial view).

In some embodiments, the registration sensor is a stylus configured to be manually operated. In some embodiments, the stylus comprises a 5DOF sensor at its distal end.

In some embodiments, the insertion device is a gastroenteral tube.

In some embodiments, the at least three anatomic locations marked by the registration sensor comprise the xiphoid process and an area in proximity to the left and right clavicular. In some embodiments, the at least three anatomic locations marked by the registration sensor comprise the xiphoid process and an area in proximity to the left and right shoulder.

In some embodiments, the determining/estimating the orientation of the subject comprises defining a first vector between the xiphoid process and the left clavicular and a second vector between the xiphoid process and the right clavicular.

In some embodiments, the position of the suprasternal notch is calculated based on the marked left and right clavicular.

In some embodiments, the registration sensor is configured to mark at least four anatomic location, wherein the fourth anatomic location comprises the patient's suprasternal notch.

In some embodiments, the at least four anatomic locations include the suprasternal notch, and the xiphoid process. In some embodiments, a frontal view display of the path of the gastroenteral tube relative to the suprasternal notch and xiphoid process is indicative of successful insertion. In some

3

embodiments, the location where the displayed path (in frontal view) of the gastroenteral tube crosses an axis between the suprasternal notch and xiphoid process is indicative of a successful insertion. In some embodiments, the actual shape of the displayed path of the gastroenteral tube is indicative of a successful insertion.

According to some embodiments, the processing circuitry may be configured to determine/calculate/draw, within the coordinate system, a first vector (v1) extending between the xiphoid process and the left clavicle and a second vector (v2) extending between the xiphoid process and the right clavicle, based on the registration of the xiphoid process and the left and right clavicles by the registration sensor. According to some embodiments, the processing circuitry may be configured to determine/calculate/draw a third vector (v3) based on the first and second vectors (v1 and v2), such as the cross product of v1 with v2, the third vector being indicative of the position, orientation and/or posture of the subject.

According to some embodiments, the system is configured to determine the path of the gastroenteral insertion tube without utilizing additional sensors.

According to some embodiments, there is provided a method for guiding an insertion device, the method comprising: utilizing an external electromagnetic field generator, applying an electromagnetic field covering a treatment area; utilizing a registration sensor, marking at least three anatomic locations on the subject's torso; and utilizing a processing circuitry to: calculate an orientation of the subject relative to the field generator based on the three anatomic locations marked by the registration sensor; load a pre-defined anatomical map representing a torso; aligning the map based on positions corresponding to the suprasternal notch or an area in proximity thereof and an anatomic location below the suprasternal notch, such as, but not limited to, the xiphoid process or the navel, based on the at least three anatomic locations marked by the registration sensor; inserting a gastroenteral tube into the patient, the gastroenteral tube comprising a tip sensor configured to sense its location relative to the electromagnetic field generator; and showing on the map a path of the gastroenteral tube insertion; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the gastroenteral tube, independent of the subject's movement and independent of deviations in the position and/or orientation of said field generator.

In some embodiments, the at least three anatomic locations marked by the registration sensor comprise the xiphoid process and the left and right clavicles.

In some embodiments, the determining/estimating the orientation of the subject comprises defining a first vector between the xiphoid process and the left clavicle and a second vector between the xiphoid process and the right clavicle.

In some embodiments, the position of the suprasternal notch is calculated based on the marked left and right clavicles.

In some embodiments, the registration sensor is configured to mark at least four anatomic locations, wherein the fourth anatomic location comprises the patient's suprasternal notch.

According to some embodiments, calculating a position, orientation and/or posture of the subject may include determining/calculating/drawing within the coordinate system, a first vector (v1) extending between the xiphoid process and the left clavicle and a second vector (v2) extending

4

between the xiphoid process and the right clavicle, based on the registration of the xiphoid process and the left and right clavicles by the registration sensor. According to some embodiments, the processing circuitry may be configured to determine/calculate/draw a third vector (v3) based on the first and second vectors (v1 and v2), for example by calculating the cross-product of v1 with v2, the third vector being indicative of the position, orientation and/or posture of the subject.

In some embodiments, the method further comprises the step of displaying the path of the distal tip section of the insertion device on the anatomical map and thus facilitates determination of a successful medical procedure.

In some embodiments, the positioning of the reference sensor comprises positioning thereof on a side of the patient's torso.

In some embodiments, the registration sensor is a manually operated stylus.

In some embodiments, the anatomical map shows a frontal upper view of the subject. In some embodiments, the anatomical map shows a side view of the subject. In some embodiments, the anatomical map shows an axial view of the subject.

In some embodiments, a path-display of the gastroenteral tube relative to the first and second anatomic locations is indicative of a successful insertion. In some embodiments, the location where the displayed path of the gastroenteral tube crosses an axis between the first and second anatomic locations is indicative of a successful insertion. In some embodiments, the shape of the displayed path of the gastroenteral tube is indicative of a successful insertion.

In some embodiments, the electromagnetic field generator is not in any physical contact with the patient. In some embodiments, the electromagnetic field generator is designed not to be in physical contact with the patient.

In some embodiments, the processor/processing circuitry may include two units or two sub-units. The first is configured to control the entire tracking system (e.g., operate the field generator, read signals obtained from the registration sensor and optionally also the reference sensor, and calculate a position and orientation thereof relative to the field generator). The second is configured to receive the calculated position and orientation information from the first processor, and use this information to generate an anatomical map representing the torso of the subject and at least the first and second anatomic locations, and to allow visualization on the anatomical map of a position, orientation and/or path of a tip sensor with respect to the first and second anatomic locations, independent of the subject's movement and independent of deviations in the position and/or orientation of the field generator.

According to some embodiments, the tip sensor is positioned at or adjacent to the distal end of the tube/catheter/stylet and aids in determining the position and orientation of the distal end of the tube/catheter/stylet in a patient.

According to some embodiments, the path of the gastroenteral insertion tube may be determined without utilizing additional sensors.

According to some embodiments, there is provided an insertion device positioning guidance system comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; the electromagnetic field generator being external to the patient; a registration sensor configured to mark a suprasternal notch and a xiphoid process on the patient's torso; a gastroenteral tube comprising a tip sensor configured to sense its location relative to the electromagnetic field generator; and a pro-

5

cessing circuitry configured to: calculate an orientation of the subject relative to the field generator based on the suprasternal notch and the xiphoid process marked by the registration sensor, wherein calculating the orientation comprises determining and optionally correcting an angle of the registration sensor relative to the patient's torso; load a predefined anatomical map representing a torso; aligning the map based on the marked suprasternal notch and xiphoid process, and showing on the map a path of the gastroenteral tube insertion; wherein the path is generated according to changes in the strength of the electromagnetic field sensed by the tip sensor's during the insertion of the gastroenteral tube, while minimizing sensitivity to the subject's movement and/or deviations in the position and/or orientation of the field generator.

According to some embodiments, the system is configured to determine the path of the gastroenteral insertion tube without utilizing additional sensors

According to some embodiments, the system further comprises a reference sensor configured to be positioned, within the treatment area, on the subject's torso, the reference sensor configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator;

In some embodiments, the reference sensor is configured to be positioned on a side of the patient's torso.

In some embodiments, the anatomical map shows a frontal upper view of the subject essentially parallel to the patient's posture. In some embodiments, the anatomical map shows a side view of the subject. In some embodiments, the anatomical map shows an axial view of the subject.

In some embodiments, the system further comprises a monitor configured to display the anatomical map.

According to some embodiments, the term "anatomical map" as disclosed herein, may refer to one or more schematic maps, one or more 2D anatomical maps, one or more 3D anatomical maps, or any combination thereof. According to some embodiments, the term "anatomical map" as disclosed herein, may refer to a group of maps (e.g., 2, 3, 4 or more), each representing a different view (for example, frontal view, frontal upper view, side view, axial view).

In some embodiments, the registration sensor is a stylus configured to be manually operated. In some embodiments, the stylus may include a 5DOF sensor at its distal end.

The systems and methods described herein may be applied, according to some embodiments, using sensors (e.g., position sensors) such as magnetic field sensors, impedance-based sensors or ultrasonic sensors. According to some embodiments, the position sensor (e.g., tip sensor) may refer to an element mounted on a catheter/tube/stylet, which causes the processing circuitry to receive signals indicative of the coordinates of the element. The position sensor may include a receiver, which generates a position signal to a processing circuitry/control unit based on energy received by the sensor (for example, from the field generator). According to some embodiments, the communication between the sensors and the processing unit may be wireless.

More details and features of the current invention and its embodiments may be found in the description and the attached drawings.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. Although methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present invention, suitable methods and materials are described below. In case of conflict, the

6

patent specification, including definitions, will control. In addition, the materials, methods, and examples are illustrative only and not intended to be limiting.

BRIEF DESCRIPTION OF THE FIGURES

Exemplary embodiments are illustrated in referenced figures. Dimensions of components and features shown in the figures are generally chosen for convenience and clarity of presentation and are not necessarily shown to scale. The figures are listed below.

FIG. 1 is a block diagram of an insertion device positioning guidance system, in accordance with some embodiments;

FIG. 2A schematically illustrates an insertion device positioning guidance system in a hospital setting, in accordance with some embodiments;

FIG. 2B shows an enlarged portion of the illustration of FIG. 2A, in accordance with some embodiments;

FIG. 2C shows a side view of the illustration of FIG. 2A, in accordance with some embodiments;

FIG. 2D-2E schematically illustrate an insertion device positioning guidance system in a hospital setting, showing anatomic locations marked using a stylus, and a reference sensor;

FIG. 3A shows a view of a "live" display of placement of an insertion device, in accordance with some embodiments;

FIG. 3B shows a view of a "playback" display of placement of an insertion device, in accordance with some embodiments; and

FIG. 4 is a flow chart of the steps of a method for guiding the position of an insertion medical device, in accordance with some embodiments;

FIG. 5 schematically shows vectors utilized for calculating an angle of a registration sensor utilized for marking of the suprasternal notch and the xiphoid process on a subject's torso and the subject's torso;

FIG. 6A and FIG. 6B schematically shows left right correction of an insertion path based on the angle calculated as described with regards to FIG. 5.

DETAILED DESCRIPTION OF THE INVENTION

Disclosed herein is a system and method for guiding insertion of an insertable medical device (e.g., a tube, such as a feeding tube). The disclosed system may be used as an insertion device positioning guidance system. The system may be used to track and indicate, in real time, the location of an insertion medical device during the insertion process. As one example, the system may track and indicate the location of a tip of a feeding tube as it is being inserted into the body of a subject. Advantageously, this makes the insertion procedure considerably easier and safer, ensuring that the tube is inserted at a correct location.

According to some embodiments, there is provided an insertion device positioning guidance system comprising: an electromagnetic field generator configured to generate an electromagnetic field covering a treatment area; a reference sensor configured to be positioned, within the treatment area, on the subject's torso, the reference sensor is configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to said field generator; a registration sensor configured to mark at least four anatomic locations relative to the reference coordinate system; and processing circuitry configured to operate said field generator, read signals obtained from said refer-

ence sensor and said registration sensor, calculate a position and orientation thereof relative to said field generator, generate an anatomical map representing the torso of the subject and at least a first and a second of the at least four anatomic locations, said processor/processing circuitry is further configured to facilitate visualization on the anatomical map of a position, orientation and path of a tip sensor, located in a distal tip section of the insertion device, with respect to the first and second anatomic locations, independent of the subject's movement and independent of deviations in the position and/or orientation of said field generator, thus determination of a successful medical procedure is facilitated. Optionally, the system further includes a monitor configured to display the map.

A reference coordinate system representing the position and orientation of the subject's torso relative to the field generator may be indicated by a reference sensor configured to be positioned, within the treatment area, on a subject's torso. The reference sensor may be positioned on a side of the patient's torso, such that the anatomical map further depicts a body contour of the subject.

The first and the second anatomic locations may be indicated by a registration sensor configured to mark at least the first and the second anatomic locations relative to the reference coordinate system. Optionally, the registration sensor is a stylus configured to be manually operated. Optionally, the first anatomic location is the suprasternal notch and the second anatomic location is the xiphoid process, and a path display of the gastroenteral tube relative to the first and second anatomic locations is indicative of a successful insertion.

The third and fourth anatomic locations may be the left and right clavicae. According to some embodiments, the left and right clavicae along with the xiphoid process may be used for calculating a position, orientation and/or posture of the subject. According to some embodiments, calculating a position, orientation and/or posture of the subject may include determining/calculating/drawing within the coordinate system, a first vector (v1) extending between the xiphoid process and the left clavica and a second vector (v2) extending between the xiphoid process and the right clavica, based on the registration of the xiphoid process and the left and right clavicae by the registration sensor. According to some embodiments, the processing circuitry may be configured to determine/calculate/draw a third vector (v3) based on the first and second vectors (v1 and v2), for example by calculating the cross product of v1 with v2, the third vector being indicative of the position, orientation and/or posture of the subject.

The electromagnetic field generator may be static throughout a duration of a procedure for placing a tube within a body of a subject. In such cases, a region covered by the electromagnetic field is static/constant throughout the duration of a procedure for placing a tube within a body of a subject. Advantageously, the static electromagnetic field may contribute to the accuracy of the display.

The anatomical map may show a frontal upper view of the subject and/or a side view of the subject and/or an axial view of the subject.

One example of hardware suitable for use as the above-mentioned electromagnetic tracking system, including the electromagnetic field generator and one or more of the sensors, is the Aurora® system by Northern Digital Inc., of Ontario, Canada.

Throughout the following description, similar elements of different embodiments of the device are referenced by element numbers differing by integer multiples of 100. For

example, an electromagnetic field generator of FIG. 1 is referenced by the number 102, and an electromagnetic field generator of FIG. 2, which corresponds to electromagnetic field generator 102 of FIG. 1, is referenced by the number 202.

Reference is now made to FIG. 1 which is a block diagram of an insertion device positioning guidance system 100. System 100 includes an electromagnetic field generator 102 configured to generate an electromagnetic field 103a covering at least a region of interest 103b (e.g., a treatment area such as a patient's torso), a plurality of electromagnetic sensors, such as sensors 104 and 106, to indicate a position of a tip sensor (located in a distal tip section of the insertion device) on an anatomical map (FIG. 3A-3B) of the region of interest 103b (typically the subject's torso). System 100 further includes a processor 110 configured to operate said field generator, read signals obtained from sensors 104 and 106, and to generate an anatomical map representing the torso of the subject. Processor 110 is configured to facilitate visualization on the anatomical map of a position and path of the tip sensor, on the map, independent of the subject's movement and independent of deviations in the position and/or orientation of field generator 102. System 100 further includes a monitor 112 operatively connected to processor 110 and configured to display, on the anatomical map, the positions of and/or the path of the insertion device tip during its insertion. In some embodiments, monitor 112 may be integrated with processor 110, such as in the case of an all-in-one computer. A determination of a successful medical procedure (for example, an insertion of a feeding tube to the stomach as opposed to the lungs) is thus possible.

Sensor 104 is typically a reference sensor configured to be positioned on a subject's torso. Reference sensor 104 is configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator. Optionally, reference sensor 104 may be attached to the skin of the patient, for example on the side of a patient's torso such as beneath the patient's armpit. In such cases, the anatomical map further depicts a body contour of the subject. Reference sensor 104 may be, for example, a 6-DOF electromagnetic sensor, capable of determining 6 axes of its location (XYZ axes) and attitude (roll, yaw, and pitch) with respect to field generator 102.

Sensor 106 is typically a registration sensor configured to be positioned on and/or to mark at least four anatomic (thoracic) locations over the subject's body (e.g. the subject's torso). Different anatomical locations may be marked depending on the type of procedure used, the type of insertion medical device, etc. The marking of the anatomic location may be physical, such as attaching a marker/fiducial (such as a sticker). Alternatively, the marking of the anatomic location may be virtual, such as registering a virtual marker/fiducial. The marking, in accordance with embodiments, may facilitate identification or designation of an anatomical location within or on a subject's body such as, in a non-limiting example, a subject's suprasternal notch, and a subject's xiphoid process.

Optionally, registration sensor 106 is a stylus sensor having a 3 DOF sensor on its distal tip, the stylus configured to be manually operated to mark at least three anatomic location over the subject's body identified by the operator of the stylus. The marking may be made, merely as an example, by indicating to the software (for example, but not limited to) by pressing a GUI button or voice activation) once stylus sensor 106 is positioned over the desired point on the patient's body. The marking may be communicated to and registered by processor 110.

System **100** is configured to work in conjunction with an insertion medical device (not shown), such as a feeding tube. The insertion medical device may include one or more sensors to allow its tracking within region of interest **103b**. Preferably, the sensor is located at the tip of the insertion medical device. In such case, processor **110** and monitor **112** are configured to compute and display position and/or advancement of the tip of the insertion medical device between the designated anatomical locations leading to the insertion site/target area.

According to some embodiments, as used herein the terms “insertion device” and “insertion medical device” may refer to any device/tool adapted for insertion into a body. The insertion device may be any medical insertion device or a medical surgical device. Non-limiting examples of insertion medical devices include, feeding tubes, such as gastroenteral tubes (for example, nasogastric feeding tubes), endotracheal tube, tracheostomy tube, stomach tube, catheter tubes or cricothyrotomy tube. Other examples of insertion devices are well known in the art.

According to some embodiments, the terms “processing circuitry” and “processor” may be used interchangeably.

In some embodiments, the insertion device is a tube. In some embodiments, the tube is a feeding tube. In some embodiments, the tube is a gastro/enteral feeding tube, such as, but not limited to, a nasogastric feeding tube or a nasogastric feeding tube. According to some embodiments, the feeding tube may have disposed therein and/or thereon an electromagnetic sensor, for example at its distal end.

Reference is now made to FIG. 2A-2E which schematically illustrate an insertion device positioning guidance system **200** in a hospital setting, in accordance with some embodiments. FIG. 2A schematically illustrates an insertion device positioning guidance system in a hospital setting, in accordance with some embodiments, FIG. 2B shows an enlarged portion of the illustration of FIG. 2A, in accordance with some embodiments, FIG. 2C shows a side view of the illustration of FIG. 2A, in accordance with some embodiments and FIG. 2D-2E schematically illustrate an insertion device positioning guidance system in a hospital setting, showing anatomic locations marked using a stylus, reference sensor (as shown in FIG. 2A-2C) located at different locations, in accordance with some embodiments.

Similar to system **100** of FIG. 1, system **200** includes an electromagnetic field generator **202**. System **200** is configured to work in conjunction with an insertion medical device (not shown) which may include one or more electromagnetic sensors configured to sense and/or interfere with the electromagnetic field generated by field generator **202**. Optionally, monitor **212** of system **200** is integrated with a computer, which corresponds to or includes processor **110** of FIG. 1.

According to some embodiments, electromagnetic field generator **202** may be positioned at such angle and position with respect to the patient, as to enable the generated electromagnetic field to cover the external and internal working area, or in other words, the entire upper torso (at least from the nose area to the duodenum area). System **200** further includes a reference sensor **204**, and stylus sensor **206** configured to be covered by the field produced by field generator **202**, when in use. The tip sensor of the feeding tube is configured to move inside the digestive system, and its path can thus be traced. Reference sensor **204** may be attached to and/or on the skin of the patient, for example beneath the patient's armpit. Suitable means for attachment of the sensor are well known in the art such as, for example, stickers, medical glue and the like. Reference sensor **204**

may serve to detect location (XYZ axes) and attitude (roll, yaw, and pitch) of the patient with respect to field generator **202**, based on the electromagnetic field (not shown) emitted by field generator **202**.

Stylus sensor **206** may be manually operated to mark at least three anatomic locations over the patient's skin. For example, FIG. 2D and FIG. 2E show the marking of four such anatomic locations (indicated as **206a**, **206b**, **206c** and **206d** in these figures) on the patient's chest. Anatomic location **206a** is marked over the suprasternal notch, anatomic location **206b** is marked over the xiphoid process and anatomic locations **206c** and **206d** are marked in the area of the left and right clavicular, respectively. The marking may be communicated to and registered by the computer.

Alternatively, only three anatomic locations may be marked, namely anatomic location **206b** marking the xiphoid process and anatomic locations **206c** and **206d** marking the left and right clavicular. The position of suprasternal notch **206a** may then be calculated from anatomic locations **206c** and **206d** (e.g., as their mid-point).

Optionally, the computer receives signals from reference sensor **204**, and optionally also from registration sensor **206** and then aligns an anatomical map representative of the subject's torso according to the marked anatomic locations and the signal of the registration sensor; thereafter the medical procedure can begin. In the exemplary case of guiding the insertion of a feeding tube, the tip of the feeding tube is equipped with a sensor. Optionally, the computer receives the actual position and orientation of the sensors from a second processor that receives the signals and calculates the sensors' locations. Optionally, the computer receives the actual position and orientation from a second processor that receives the signals from the sensors and calculates their physical location.

System **200** is operated as follows:

The electromagnetic field generator **202** is activated to apply an electromagnetic field to the treatment area, covering the subject's torso;

Reference sensor **204** is positioned within the treatment area, on a subject's torso, preferably on the side of the torso. Reference sensor **204** defines a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator;

Registration sensor **206** is used to mark at least three anatomic locations on the subject's torso, preferably including the suprasternal notch, the xiphoid process and the left and right clavicular;

Utilizing a processor, aligning an anatomical map representing the torso to the at anatomic locations, e.g. **206a** and **206b** derived from the at least three anatomic locations marked by registration sensor **206** and displaying on monitor **212** the anatomical map and the position and path of the tip sensor (of the feeding tube). The path of the tip sensor may be displayed with respect to anatomic locations **206a** and **206b** and/or with respect to a longitudinal axis passing between anatomic locations **206a** and **206b** and along the center of the torso.

Reference is now made to FIG. 3A, which shows a view of a “live” display of placement of an insertion device, in accordance with some embodiments and to FIG. 3B, which shows a view of a “playback” display of placement of an insertion device, in accordance with some embodiments. Such displays may be presented on a monitor such as monitor **212**. The left corner includes general information and patient's details, and in the display of FIG. 3B, also playback controls.

11

The tip's path is schematically drawn, enabling the caregiver to visualize the entire insertion path of the tube, until it reaches the desired location. Optionally, and as shown in FIG. 3A and FIG. 3B, an arrow is near the tip of the path, indicating the actual direction to which the tube is pointing. Such arrow(s) may help the user to properly insert the tube (or better understand where and to which direction the tube is moving). The displays of both FIG. 3A and FIG. 3B show three views of the patient's body: a frontal view shown at the top right side of the monitor, a lateral view shown at the bottom left side of the monitor, and an axial view shown at the bottom right side of the monitor. In some embodiments, different and/or additional views may be shown.

The caregiver inserting the insertion medical device can view the indications on monitor 212 while manually maneuvering the insertion tube into the patient's body, so as to guide it to the desired location in the body.

Reference is now made to FIG. 4, which is a flow chart of the steps of a method for guiding the position of an insertion medical device, in accordance with some embodiments. Step 420 includes application of an electromagnetic field to a treatment area. Optionally, an electromagnetic field generator, such as electromagnetic field generator 202 of FIG. 2, is positioned such that the electromagnetic field covers the treatment area. Optionally, a reference sensor, such as reference sensor 204 of FIG. 2A-2D, is positioned on a patient within the treatment area, on a subject's torso, the reference sensor configured to define a reference coordinate system representing the position and orientation of the subject's torso relative to the field generator (step 422). Optionally, the reference sensor is positioned on a side of the patient's torso such as to indicate a body contour/width of the subject. At least three anatomic locations are marked by utilizing a registration sensor (step 424). Optionally, a stylus sensor, such as stylus sensor 206 of FIG. 2A-2D may be used, the stylus configured to be manually operated to mark the anatomic locations (e.g., anatomic locations 206a-206d). Alternatively, one or more registration sensors may be positioned over the at least three anatomic locations (option not shown). Optionally, the first anatomic location is the suprasternal notch, the second anatomic location is the xiphoid process and the third and fourth anatomic locations are the left and right clavulae, respectively. According to some embodiments, the position of the suprasternal notch may be calculated from the marked left and right clavulae (e.g. the mid-point therebetween). Each of steps 420, 422 and 424, may be performed simultaneously or in an interchangeable order. An anatomical map representative of the subject's torso may then be aligned to correspond to the marked anatomic locations and optionally to the signals obtained from the reference sensor (step 426). A position and orientation of a tip sensor of the insertion device may then be displayed on the anatomical map, with respect to the first and second anatomic locations, independent of the subject's movement and independent of deviations in the position and/or orientation of the field generator (step 428). This can be accomplished when all sensors remain within the sensing volume of the field generator.

Optionally, the anatomical map shows a frontal upper view of the subject, a side view of the subject, and an axial view of the subject. Optionally, the anatomical map enables visualization of the location of an insertion device (having an electromagnetic sensor configured to sense and/or interfere with the electromagnetic field generated by the field generator) within a subject's body by computing and displaying a position of the inserted device vis-à-vis at least first and second anatomic locations 206a and 206b. Option-

12

ally, the computing may include normalizing the position based signals received from reference sensor 204 and/or anatomic location 206c and 206d marked by registration sensor 206, optionally obtained continuously and/or in real-time during the procedure.

Reference is now made to FIG. 5 which schematically illustrate vectors utilized for calculating an angle of a registration sensor (stylus), utilized for marking of the suprasternal notch and the xiphoid process on a subject's torso, and the subject's torso.

It is understood that if the registration sensor is perpendicular to the subject's torso, the orientation (head-leg and left-right) of the subject may be determined directly by the registration sensor's angle. However, if the registration sensor is tilted towards the subject's head, legs, left or right, the angle of the registration may be determined and corrected.

The calculating of the angle is founded on two preliminary assumptions:

1. The line connecting the marked points A and B is aligned with the head-legs axis $\vec{AE}$.
2. The angle α between $\vec{AB}$ and the patient back is predefined (17°).

Based on these assumptions, the up-vector can be calculated and corrected as follows:

$\vec{BC}$ represents the direction of the registration sensor and $\vec{BD}$ the vector perpendicular to the plane created from $\vec{BA}$ and $\vec{BC}$. $\vec{BC}$ is then around axis BD (to get $\vec{BC}'$) so that the angle between the patient's back plane and AB is α , thereby obtaining a corrected up-vector.

After calculating the up-vector, it may still be tilted to one of the sides (left or right).

Such left-right tilting may be corrected by the user after the placement of the gastroenteral tube has been commenced and enough 3D path data points have been collected (up to some point between the points A and B). Since the user expects that in the frontal view, the path between the data points be vertical (as the esophagus is going straight down, parallel to $\vec{AE}$), the user can choose to "realign" the path if the line obtained is not straight as seen in FIG. 6a.

In that case, the processing circuitry may be configured to iterate on different angles around the $\vec{AB}$ axis, and to recalculate the path to minimize the angle between $\vec{AB}$ and the regenerated path (so that the angle that makes the frontal path is vertical as possible). Once minimized (for example using gradient descent technique), an up-vector corrected in all directions (head-leg, right-left) is obtained is illustrated in FIG. 6b.

The present invention may be a system, a method, and/or a computer program product. The computer program product may include a computer readable storage medium (or media) having computer readable program instructions thereon for causing a processor to carry out aspects of the present invention.

The computer readable storage medium can be a tangible device that can retain and store instructions for use by an instruction execution device. The computer readable storage medium may be, for example, but is not limited to, an electronic storage device, a magnetic storage device, an optical storage device, an electromagnetic storage device, a semiconductor storage device, or any suitable combination of the foregoing. A non-exhaustive list of more specific

13

examples of the computer readable storage medium includes the following: a portable computer diskette, a hard disk, a random access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a static random access memory (SRAM), a portable compact disc read-only memory (CD-ROM), a digital versatile disk (DVD), a memory stick, a floppy disk, a mechanically encoded device having instructions recorded thereon, and any suitable combination of the foregoing. A computer readable storage medium, as used herein, is not to be construed as being transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide or other transmission media (e.g., light pulses passing through a fiber-optic cable), or electrical signals transmitted through a wire. Rather, the computer readable storage medium is a non-transient (i.e., not-volatile) medium.

Computer readable program instructions described herein can be downloaded to respective computing/processing devices from a computer readable storage medium or to an external computer or external storage device via a network, for example, the Internet, a local area network, a wide area network and/or a wireless network. The network may comprise copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and/or edge servers. A network adapter card or network interface in each computing/processing device receives computer readable program instructions from the network and forwards the computer readable program instructions for storage in a computer readable storage medium within the respective computing/processing device.

Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like, and conventional procedural programming languages, such as the "C" programming language or similar programming languages. The computer readable program instructions may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider). In some embodiments, electronic circuitry including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention.

Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems), and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the

14

flowchart illustrations and/or block diagrams, can be implemented by computer readable program instructions.

These computer readable program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computer, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the flowchart and/or block diagram block or blocks.

The computer readable program instructions may also be loaded onto a computer, other programmable data processing apparatus, or other device to cause a series of operational steps to be performed on the computer, other programmable apparatus or other device to produce a computer implemented process, such that the instructions which execute on the computer, other programmable apparatus, or other device implement the functions/acts specified in the flowchart and/or block diagram block or blocks.

The flowchart and block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods, and computer program products according to various embodiments of the present invention. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of instructions, which comprises one or more executable instructions for implementing the specified logical function(s). In some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts or carry out combinations of special purpose hardware and computer instructions.

The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

What is claimed is:

1. A guidance system for insertion of a tube, the system comprising:

a processing circuitry configured to:

calculate an orientation of the subject relative to the electromagnetic field, based on at three anatomic locations marked on a subject's torso using a stylus,

15

wherein the at least three anatomic locations comprise the xiphoid process and the left and right clavicular load an anatomical map representing the torso;

deriving a position of the suprasternal notch, based on the marked left and right clavicular;

aligning the map to the subject's torso, based on the position of the xiphoid process marked by the stylus and the derived position of the suprasternal notch, receiving signals indicative of an electromagnetic field sensed by a passive electromagnetic sensor of a tube during its insertion;

showing on the aligned map a path of the tube during its insertion, based on the received signals.

2. The system of claim 1, wherein the anatomical map shows a frontal upper view of the subject, a side view of the subject and/or an axial view of the subject.

16

3. The system of claim 1, further comprising a monitor configured to display the map.

4. The system of claim 1, wherein determining/estimating the orientation of the subject comprises defining a first vector between the xiphoid process and the left clavicle and a second vector between the xiphoid process and the right clavicle.

5. The system of claim 1, wherein the aligning comprises adjusting the width and/or length of the representative torso to the width and/or length of the subject's torso.

6. The system of claim 5, wherein the aligning further comprises indicating the position of the xiphoid process and the suprasternal notch on the adjusted representative torso.

7. The system of claim 6, wherein the aligning further comprises indicating the position of the left and right clavicular on the adjusted representative torso.

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